Agent Blueprint — End-to-End SDLC

Initiative → Epics → Stories → Design → Code → Tests → PR | 14 Agents | 6 KBs | 6 Guardrails | 9 Tools | 7 Quality Gates

Knowledge Base | Guardrail | Tool | Quality Gate | Agent | Output Artifact

📋 **INPUT: "Add domestic fund transfer capability to mobile banking"**

INCEPTION

🤖 A1: Epic Creator
IN: Initiative description
OUT: Epics with scope, goals, success criteria
kb-L1-enterprise-security | kb-L2-payments-domain
gr-L1-schema-validator | gr-L1-hallucination-check
QG: epic-quality ≥7.0
Patterns: Reflection + Grounding

🤖 A2: Feature Decomposer
IN: Epics from A1
OUT: Features with business justification
kb-L1-enterprise-security | kb-L2-payments-domain
gr-L1-schema-validator | gr-L1-hallucination-check | gr-L2-payments-terminology
QG: feature-completeness ≥7.0
Patterns: Reflection + Grounding + CoT

🤖 A3: Story Generator ⭐
IN: Features from A2
OUT: User stories with AC (Given/When/Then), data sensitivity, citations
kb-L1-enterprise-security | kb-L1-sdlc-templates | kb-L2-payments-domain
gr-L1-schema-validator | gr-L1-pii-detection | gr-L1-hallucination-check
gr-L2-payments-compliance | gr-L2-payments-terminology
QG: story-quality ≥7.0 | VERIFY: A12 ≥7.0
L1-jira-fetch-issue
Patterns: Reflection + Grounding + Citation

🤖 A4: Task Generator
IN: Stories from A3
OUT: Dev, test, security, compliance tasks
kb-L1-sdlc-templates
gr-L1-schema-validator
QG: task-completeness ≥6.0
Patterns: Reflection + Grounding

⚙️ A13: Jira Orchestrator
IN: Epics + Stories + Tasks
OUT: 📌 Jira issues created (labels: aava-generated)
L1-jira-create-issue | L1-jira-update-issue

DESIGN

🤖 A5: UX Wireframe
IN: Stories from A3
OUT: Component hierarchy, screen layouts, interaction flows
kb-L1-enterprise-architecture
gr-L1-schema-validator
QG: ux-completeness ≥7.0
Patterns: Reflection + Verification

🤖 A7: HLD Designer
IN: Stories + Features
OUT: C4 diagrams, component design, data flow, integrations
kb-L1-enterprise-architecture | kb-L1-enterprise-security | kb-L2-payments-domain
gr-L1-schema-validator | gr-L1-hallucination-check | gr-L1-consistency-check
QG: hld-quality ≥7.0 | VERIFY: A12 ≥7.0
Patterns: Reflection + CoT + Verification

🤖 A6: API Spec Generator
IN: Stories + HLD from A7
OUT: OpenAPI 3.0 YAML (endpoints, schemas, auth, errors)
kb-L1-enterprise-architecture | kb-L2-payments-api-patterns
gr-L1-schema-validator | gr-L2-payments-compliance
Patterns: Reflection + Grounding + Schema

🤖 A8: LLD Designer
IN: HLD + API spec
OUT: Class design, DB schema, sequence diagrams, error handling
kb-L1-enterprise-architecture | kb-L1-enterprise-security
gr-L1-schema-validator | gr-L1-hallucination-check | gr-L1-consistency-check
QG: lld-quality ≥7.0
Patterns: Reflection + CoT + Verification

⚙️ A14: GitHub Orchestrator
IN: UX + HLD + LLD + API spec files
OUT: 🔀 Design branch committed (4 docs)
L1-github-upload-file

CONSTRUCTION

🤖 A9: Code Generator
IN: LLD + API spec + Stories
OUT: Spring Boot services + React components (15 files, 847 lines)
kb-L1-enterprise-architecture | kb-L1-enterprise-security
gr-L1-schema-validator | gr-L1-pii-detection
QG: code-quality ≥7.0 | VERIFY: A12 ≥7.0
Patterns: Verification + Grounding

⚙️ A14: GitHub Orchestrator
IN: Code files
OUT: 🔀 Implementation branch committed
L1-github-upload-file

TESTING

🤖 A10: Scenario Writer
IN: Stories + AC + API spec
OUT: Test scenarios (functional, security, edge cases)
kb-L1-testing-standards | kb-L2-payments-domain
gr-L1-schema-validator | gr-L1-hallucination-check
QG: test-coverage ≥7.0
Patterns: Reflection + Grounding + Citation

🤖 A11: Case Writer
IN: Test scenarios from A10
OUT: Cucumber test cases (Given/When/Then)
kb-L1-testing-standards
gr-L1-schema-validator
Patterns: Reflection + Grounding

⚙️ A13: Jira/XRay Orchestrator
IN: Test cases
OUT: 🧪 24 XRay test cases linked to stories
L1-xray-create-test

▼ All artifacts ready ▼

⚙️ A14: GitHub Orchestrator → 🔀 Pull Request #142
📌 Jira: 2 epics, 12 stories, 36 tasks | 📄 Designs: HLD, LLD, OpenAPI, UX | 💻 Code: 15 files | 🧪 Tests: 24 cases
L1-github-create-pr

🔍 A12: Quality Verifier (Cross-Cutting)
IN: Any agent output + rubric
OUT: Quality score (0-10) + findings + pass/fail
Called after: A3 (stories), A7 (HLD), A9 (code)
Independent agent — different prompt, different perspective
Rubrics: story-quality, hld-quality, code-quality

🛡️ Guardrail Chain (Applied to Every Agent)
1. gr-L1-output-schema-validator → Block if schema invalid
2. gr-L1-pii-detection → Block if PII found
3. gr-L1-hallucination-check → Flag ungrounded claims
4. gr-L2-payments-compliance → Validate PSD2/SCA refs
5. gr-L2-payments-terminology → Validate domain terms
6. gr-L1-consistency-check → Flag contradictions

📚 Knowledge Bases (6)
kb-L1-enterprise-architecture → A5,A6,A7,A8,A9
kb-L1-enterprise-security → A1,A2,A3,A7,A8,A9
kb-L1-sdlc-templates → A3,A4
kb-L1-testing-standards → A10,A11
kb-L2-payments-domain → A1,A2,A3,A7,A10
kb-L2-payments-api-patterns → A6

14	9	6	6	7	16
Agents	Tools	Knowledge Bases	Guardrails	Quality Gates	Workflow Steps